

Multi-Modal Gemini Chatbot

Technical Documentation & Implementation Guide

Problem Statement

Modern AI workflows are plagued by tool fragmentation. Users must navigate between specialized models for text, vision analysis, and image synthesis. This leads to **context drift**, where the insights gained from an image cannot be easily synthesized into a textual report, and vice-versa, without significant manual effort and copy-pasting across disparate platforms.

Methodology & Architecture

The system utilizes a **Synchronous Multimodal Backbone**:

- **Data Ingestion:** Streamlit's file-buffer system handles multipart/form-data uploads, ensuring low-latency transfers to the model.
- **Core Reasoning (Gemini 3 Flash):** Employs native multimodal tokens. Instead of converting images to text (OCR), the model perceives pixels directly, allowing for spatial reasoning and style analysis.
- **Image Generation (Nano Banana 2):** Integrated as an output modality. Using `gemini-3.1-flash-image-preview`, the system can generate visual assets in real-time within the same conversation thread.
- **Secure Tunnelling:** Cloudflare Warp/Tunnels create an encrypted bridge (Port 8501) to a public `.trycloudflare.com` URL, enabling remote access without server configuration.

Requirements Checklist

Google GenAI SDK

Streamlit 1.30+

Python 3.10+

Cloudflare CLI

- **API Access:** A valid API Key from Google AI Studio with permissions for Gemini 3.1 models.
- **Environment:** Python environment with Pillow for image manipulation and io for byte-stream handling.

Professional Usage Instructions

Local Deployment

```
pip install google-genai streamlit Pillow
streamlit run app.py
```

Cloudflare Integration

```
cloudflared tunnel --url http://localhost:8501
```